SIMATS ENGINEERING

ASSESSMENT TOOL 3: Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

Code of Conduct

I SATHYAM SHARMA (192526028 2) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T. Sathyam

Assessment Tasks

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

Here are 10 additional technical presentation topics in a similar style, suitable for a 6–8 minute presentation with definitions, architecture, industry relevance, and a practical example:

1. **Containerization and Cloud-Native Applications: Transforming Modern Software Deployment**
   - Explore containers, container architecture, benefits, and a real-world deployment example.
2. **Serverless Computing: Reducing Infrastructure Management in the Cloud**
   - Discuss Function-as-a-Service (FaaS), architecture, business advantages, and practical use cases.
3. **Edge Computing vs Cloud Computing: Choosing the Right Computing Paradigm**
   - Compare architectures, applications, and industry adoption scenarios.
4. **Microservices Architecture in Cloud Environments**
   - Explain microservices design, benefits over monolithic systems, and industry implementations.
5. **Cloud Security Fundamentals: Protecting Data and Applications**
   - Cover security models, shared responsibility architecture, and real-world security practices.
6. **Multi-Cloud and Hybrid Cloud Strategies for Enterprise IT**
   - Analyze deployment models, business benefits, challenges, and practical examples.

TECHNICAL PRESENTATION

Team no. 3

12. Big Data Analytics in Cloud Computing

CSA1519 – Cloud Computing and Big Data Analytics

Guided By
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Abstract

- Big data → massive, diverse and rapidly generated datasets
- Cloud platforms → scalable storage and on-demand processing
- Distributed systems → process data in parallel across clusters
- Analytics → descriptive, predictive and real-time insights
- Focus → architecture, platforms, business value and applications

Aim and Objective

- Aim → explore big data analytics using cloud computing
- Understand cloud-based data processing architectures
- Examine storage, distributed processing and analytics platforms
- Identify business value, scalability and cost benefits
- Explore security, governance and data management challenges
- Study real-world applications across industries

Introduction

- Big data is characterized by volume, velocity and variety
- Cloud computing provides resources over the internet on demand
- Cloud storage supports large, growing datasets
- Distributed processing handles data in parallel
- Analytics converts raw data into actionable insights

Existing System

- Traditional systems often rely on fixed on-premises capacity
- Large datasets may require costly hardware upgrades
- Scaling storage and compute can take time
- Separate tools can create data silos and complex workflows
- Cloud platforms offer elastic resources and managed services
- Challenges → cost control, security and vendor dependence

Overall System Architecture Design

- Sources → applications, business systems, devices and logs
- Ingestion layer → batch loads and real-time streams
- Storage layer → cloud object store / data lake / warehouse
- Processing layer → distributed compute and data preparation
- Analytics layer → BI dashboards, ML models and reports
- Security layer → IAM, encryption, governance and monitoring

Core Technical Explanation

- Data sources → apps, sensors, transactions and social media
- Ingestion → batch or streaming data enters the cloud
- Storage → data lakes, object stores and cloud warehouses
- Processing → distributed engines clean and transform data
- Analytics → dashboards, machine learning and prediction
- Key concerns → quality, privacy, security and governance

Proposed Methodology and Algorithm



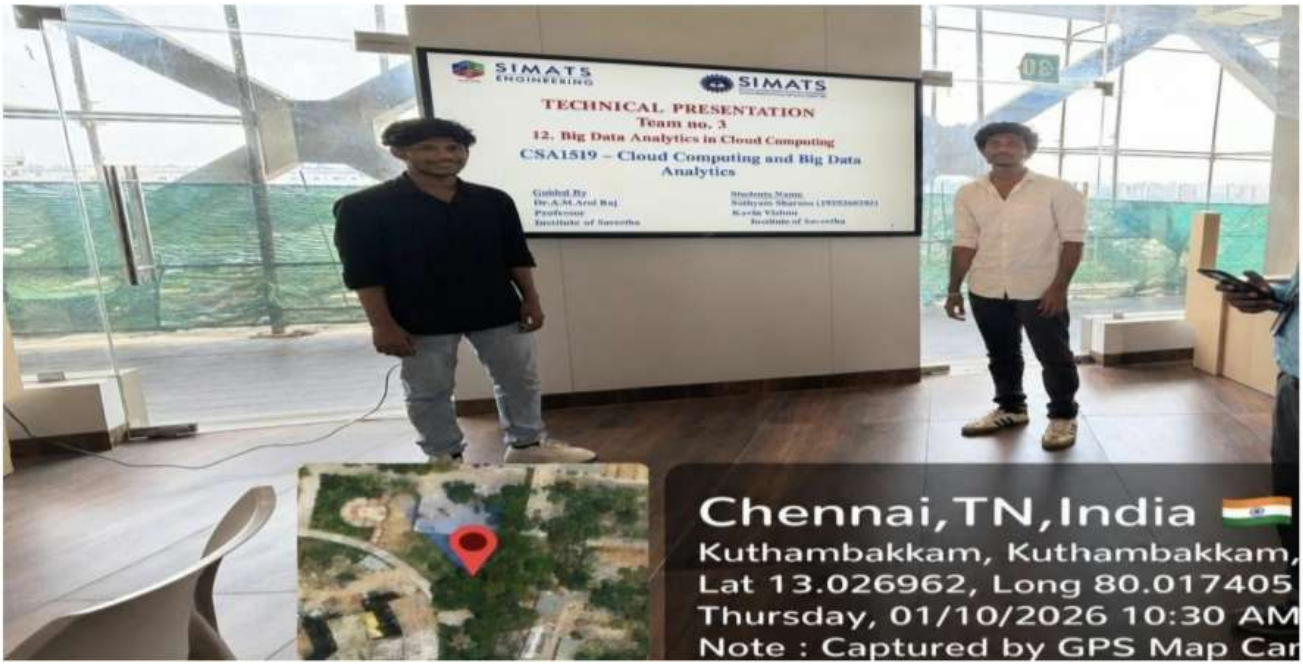
1. Identify business question and required data
2. Ingest data using batch or streaming pipelines
3. Validate, clean and transform incoming datasets
4. Store data in a scalable and governed cloud repository
5. Process data in parallel and run analytics models

Real-world use case in cloud computing

Retail → demand forecasting, recommendations and customer trends
Healthcare → population insights and clinical data analysis
Banking → fraud detection and risk analytics
Manufacturing → predictive maintenance and quality monitoring
Transport → route optimization and fleet analytics
Smart cities → traffic, energy and environmental monitoring

Conclusion

Cloud computing makes large-scale analytics more accessible
Distributed processing supports fast, scalable data workloads
Cloud platforms integrate storage, processing and machine learning
Business value includes better decisions and operational efficiency
Security, governance and cost management remain essential
Successful projects align architecture with business needs



TECHNICAL PRESENTATION

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CLOUD COMPUTING &
BIG DATA ANALYSIS
CSA1519

Big Data analytics in cloud computing :-

Big data Analytics in cloud computing is the practical processing and analyzing large amount of data using cloud technology platforms like AWS, Microsoft Azure, Hadpop and google ~~it~~ cloud helps process data efficiently

It helps busness reduce costs improve decision-making, predict trends, and understand consumer needs

Healthcare, banking retail, manufacturing and transportation use big data Analytics for better Performance and planning

Despite its advantages, big data analytics in cloud computing faces several challenges. These include data Privacy, cybersecurity threats, poor data quality, integration difficulties and unexpected cloud costs. organizations must implement strong security measures proper access controls, data governance Politics and cost-monitoring System to overcome these challenges

Big data Analytics in cloud computing is an important technology that transforms large amounts of data into useful information. It Supports better decesion making, improves buissness Performance, and helps organizations, develop smarter and more efficient systems.